

Project Proposal

CMPT 310 - Summer 2026

Alethia - AI Medical Symptom-to-Condition Classifier

Predicting likely disease from patient-reported symptoms using supervised classification

1. Group Information

Group Number: Group 09

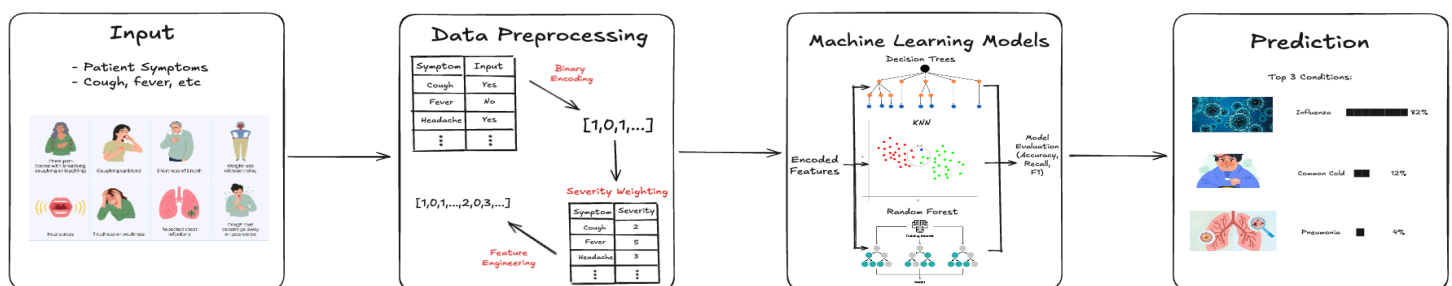
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2. Project Idea

We will build a machine learning system that accepts a set of patient-reported symptoms as input and predicts the most likely disease or medical condition as output. Given a binary feature vector representing the presence or absence of symptoms - such as fever, cough, fatigue, or chest pain - the system will classify the input into one of 41 disease categories, including influenza, pneumonia, diabetes, and migraine. The model will also return the top-3 most likely conditions with associated confidence scores, allowing users to understand uncertainty in each prediction.

Unlike simply searching symptoms on Google - which returns a flood of unranked, often alarming possibilities with no sense of relative likelihood - our system applies a trained classifier to produce a ranked, confidence-weighted list of conditions based on patterns learned from thousands of real patient cases. This makes the output more structured, more interpretable.

Our primary model will be a Decision Tree classifier, chosen for its interpretability: every prediction comes with a human-readable explanation tracing which symptom combinations led to a given diagnosis. We will benchmark it against a K-Nearest Neighbors (KNN) classifier and a Random Forest to evaluate whether ensemble methods meaningfully improve accuracy. As a stretch goal, we will build a simple web interface where users select symptoms via checkboxes and see predictions update live.



System Input/Output:

- **Input:** Binary symptom vector (~131 symptom flags), e.g. fever=1, cough=1, rash=0
- **Model:** Decision Tree (primary), KNN and Random Forest (baselines), trained on ~4,920 patient cases
- **Output:** Top predicted condition + top-3 ranked conditions with confidence scores per class

3. Tools and Resources

Libraries:

- **pandas, NumPy**: data loading, cleaning, and binary encoding of symptom columns
- **scikit-learn**: DecisionTreeClassifier, KNeighborsClassifier, RandomForestClassifier, cross-validation, classification report, confusion matrix.
- **matplotlib / seaborn**: confusion matrix heatmap, feature importance bar chart, decision tree visualization
- **Streamlit (stretch goal)**: lightweight interactive web UI for the live symptom checklist demo

Data Sources:

- **Disease Symptom Prediction dataset (Kaggle)**: 4,920 patient records across 41 disease classes, each encoded as up to 17 symptom columns. Available as a clean CSV download; no scraping or data generation required.
- **Symptom Severity dataset (Kaggle)**: maps each symptom to a severity weight (1–7), used as additional features in our improved model.

No simulator is required. Our data is real-world, collected from clinical records, publicly available, and pre-cleaned. We will not need to generate a synthetic dataset.

4. Project Plan / Timeline

Phase	Dates	Key Tasks/Activities
Setup & Data	Jun 10–18	Download and inspect dataset, verify class balance, binary-encode symptom columns, perform 80/20 stratified train/test split.
Feature Engineering	Jun 19–27	Encode symptom severity weights, analyze class distribution, identify and handle missing or inconsistent labels.
Milestone 1 (Jul 1)	Jun 28–Jul 1	KNN baseline trained and evaluated, accuracy/precision/recall reported, data pipeline complete and reproducible.
Model Improvement	Jul 2–15	Train Decision Tree with hyperparameter tuning, visualize tree structure, train Random Forest, plot feature importances.
Top-3 & Explainability	Jul 16–24	Extend output to return top-3 conditions with confidence, display decision path, add severity weights.
Milestone 2 (Jul 29)	Jul 25–29	Full system completes with models compared, evaluation report written, visualizations finalized, stretch: Streamlit UI deployed.
Write-up & Demo	Jul 30–Aug 12	Final write-up, code clean-up, How-To Guide, prepare demo.

5. Minimal Viable System

The simplest working version of our system is a Decision Tree classifier that accepts a binary symptom vector - manually entered as a Python list - and returns a single predicted condition. This requires only the CSV dataset, one-hot encoding of symptom columns, and a trained sklearn DecisionTreeClassifier, achievable within the first week of the project.

Core functionality prioritized first:

- CSV loaded and symptom columns binary-encoded into a feature matrix.
- Decision Tree trained and evaluated with accuracy and a confusion matrix.

- Single prediction function: input symptom list $\rightarrow$ output predicted condition.
- Tree visualized with plot tree, showing the decision path for any prediction.

Even in this minimal form, the system is useful and insightful: it provides a transparent, explainable prediction that shows exactly which symptom combinations drive each diagnosis. The decision tree visualization alone makes for a compelling demo, and the baseline accuracy gives a concrete benchmark to improve upon with Random Forest and additional features.